e) There is a faculty member who has never been asked a question by a student: $\exists x(F(x)\wedge \forall y(S(y)\Rightarrow \neg A(y,x)))$

f) Some student has asked every faculty member a question:

$\exists x(S(x)\wedge \forall y(F(y)\Rightarrow A(x,y)))$??

g) There is a faculty member who has asked every other faculty member a question: $\exists x(F(x)\wedge \forall y((x\neq y)\wedge F(y)\Rightarrow A(x,y)))$

h) Some student has never been asked a question by a faculty member:

$\exists x(S(x)\wedge \forall y(F(y)\Rightarrow \neg A(y,x)))$

12) Let $I(x)$ be the statement "x has an Internet connection" and $C(x,y)$ be the statement "x and y have chatted over the Internet," where the domain for the variables x and y consists of all students in your class. Use quantifiers to express each of this statements.

a) Jerry does not have an Internet connection: $\neg I(\text{Jerry})$

b) Rachel has not chatted over the Internet with Chelsea: $\neg C(\text{Rachel}, \text{Chelsea})$.

c) Jane and Sharon have never chatted over the Internet: $\neg C(\text{Jane}, \text{Sharon})$.

d) No one in the class has chatted with Bob: $\forall x \neg C(x, \text{Bob})$

e) Sanjay has chatted with everyone except Joseph: $\neg C(\text{Sanjay}, \text{Joseph}) \wedge \forall x((x\neq \text{Joseph})\Rightarrow C(\text{Sanjay}, x))$

f) Someone in your class does not have an Internet connection:

g) Not everyone in your class has an Internet connection: $\neg\forall x\, I(x)$

$\exists x \neg I(x)$

g) Not everyone in your class has an Internet connection:

h) Exactly one student in your class has an Internet connection:

$\exists x\, I(x) \wedge \forall y((x\neq y)\Rightarrow \neg I(y))$

i) Everyone except one student in your class has an Internet connection

: $\exists x \neg I(x) \wedge \forall y((x\neq y)\Rightarrow I(y))$

j) Everyone in your class with an Internet connection has chatted